

Debugging Walkthrough: Tracing a Bug From UI to Database

Symptom: A data table rendered stale/incorrect values after an update, even though the save "succeeded."

How I traced it (UI → API → data layer):

- Confirmed the UI was sending the correct payload (network tab / request inspection).
- Checked the API response — the write succeeded, but the returned object didn't reflect the change.
- Traced to the data layer: the read was hitting a cached/earlier value, so the refetch returned old data.

Fix: Invalidated the cached query on mutation success so the table refetched fresh data. Small, contained change — no schema or API contract changes needed.

Verification: Reproduced the original bug, applied the fix, confirmed the table updated immediately, and checked that the adjacent flows sharing the same query weren't affected.

Takeaway: Most "frontend" bugs in data-heavy SaaS are actually data-flow bugs. I trace the full path before touching code.